

# Lesson 5-4 · Period 1

## Solving Radical Equations — Foundation

Klimesara-adapted · Student-centered · No Blooket

DOK 2

5 min

*notice & wonder*

**Explore & Reason · Launch problem** **A.** Solve  $3(a + 1)^2 + 2 = 11$ . Use at least two methods.

**B.** Try each method on  $3\sqrt{a + 1} + 2 = 11$ .

**C.** Which method works better with a radical? What problems arise with the other?

# Launch — Isolate–Raise–Check Rules

Algebra 2 .

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DOK 1-2

12 min

teacher models

## Isolate–Raise–Check Rules (keep on your page)

1. **Isolate** the radical on one side of the equation.
2. **Raise** both sides to the index power ( $\sqrt{\phantom{x}} \rightarrow$  square;  $\sqrt[3]{\phantom{x}} \rightarrow$  cube).
3. Solve the resulting polynomial equation.
4. **Check** each candidate in the *original* equation — reject any extraneous solutions.

**COMMON ERROR:** Squaring is NOT reversible — always substitute into the ORIGINAL

# Explore — Think-Pair-Share (33 min)

Algebra 2 .

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DOK 1-2

33 min

student work

6 items in order. Plan ~5–6 min per item. Cut #4 if pace slows.

**Try It 1 · Isolate and raise to solve**  $\sqrt{x-2} + 3 = 5$   
and  $\sqrt[3]{x-1} = 2$

**Try It 3 · Solve + check for extraneous**  $x = \sqrt{7x+8}$   
and  $x+2 = \sqrt{x+2}$

**Practice #22 + #8 · Solve radical equations**

DOK 2

5 min

*academic conversation*

**Sentence frame · Pair share** “I isolated the radical to get \_\_\_\_\_. Squaring both sides gives \_\_\_\_\_. My candidate solutions are \_\_\_\_\_. When I check in the ORIGINAL equation, \_\_\_\_\_ is valid and \_\_\_\_\_ is extraneous because \_\_\_\_\_.”

**Bridge to Period 2** Next class: equations with two radicals or a radical and a binomial.

Same Isolate–Raise–Check logic — more complex setup

# Exit Ticket — Summary Recap

Algebra 2 · Lesson 5-4 ·

Period 1

DOK 1-2

2 min

*not graded*

## Complete on your exit ticket

1. To solve a radical equation I first \_\_\_\_\_.
2. I raise both sides to power \_\_\_\_\_ because \_\_\_\_\_.
3. An extraneous solution happens when \_\_\_\_\_.